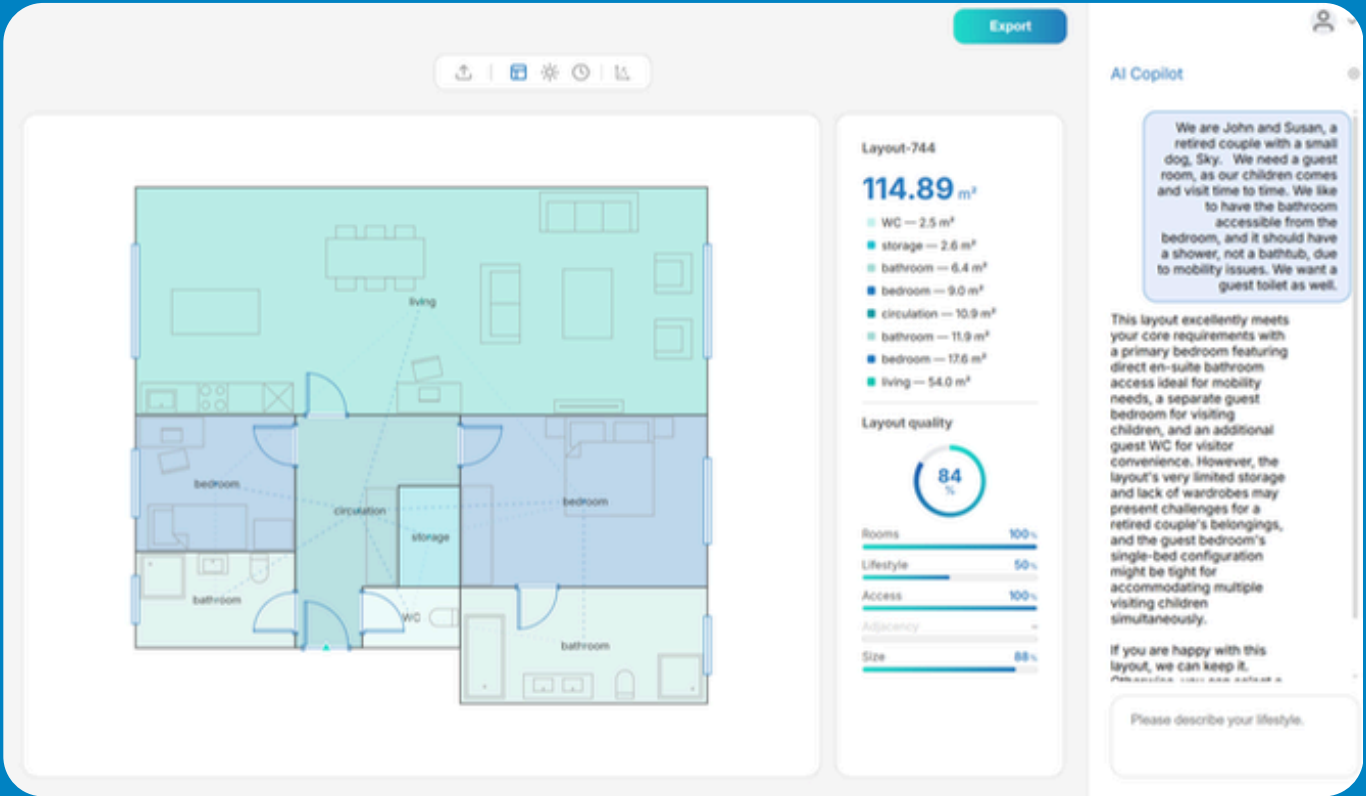


INHABIT

AI SPATIAL INTELLIGENCE COPILOT



TYPE: AI agent

YEAR: 2026

TEAM: Andrea Cutroni, Elena Corio, Symon Kipkemei, Marina Osmolovska & Giovanni Carlo Volpe

FACULTY: João M. Silva, Bao Trinh and Scott Lebow

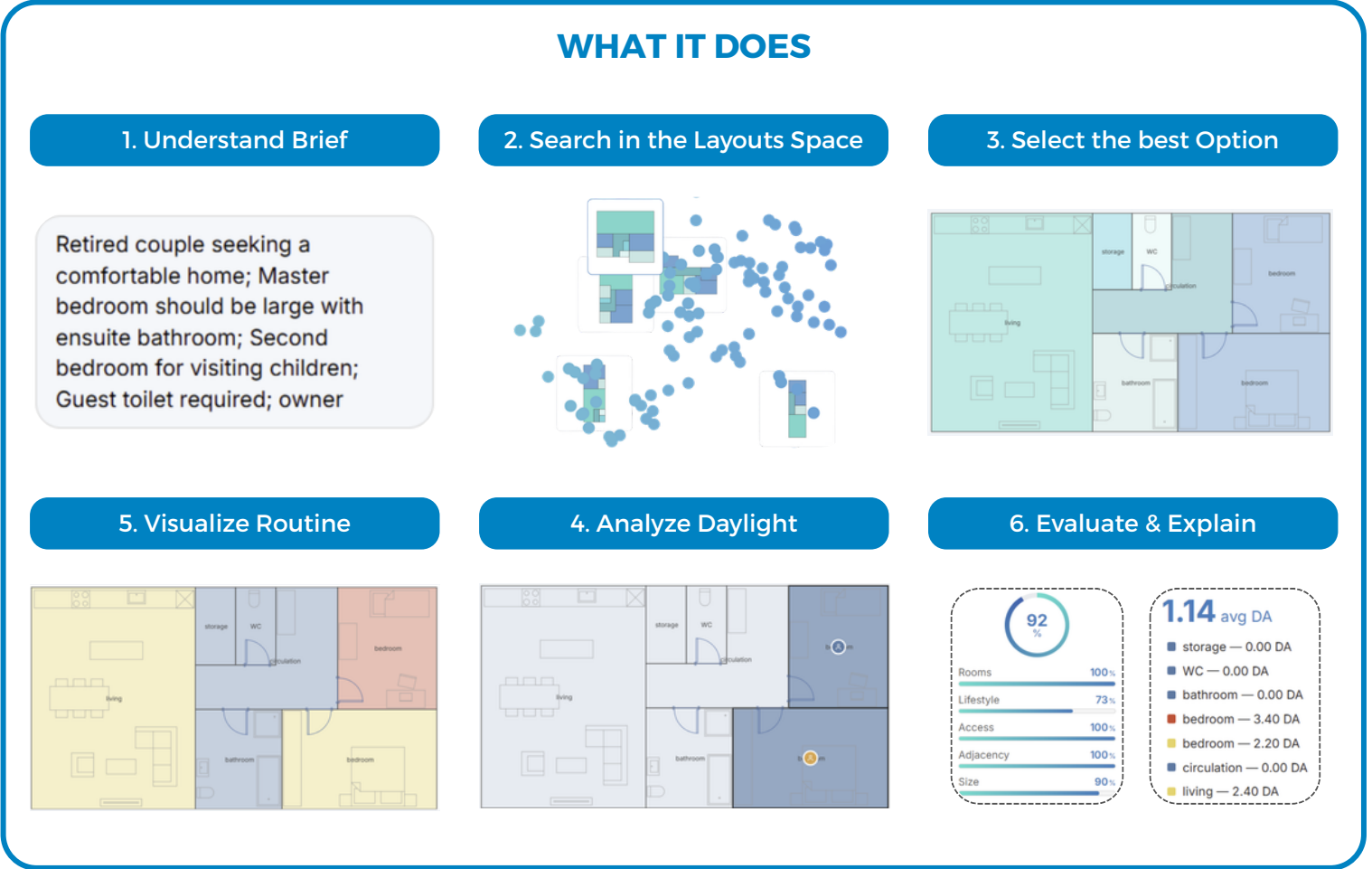
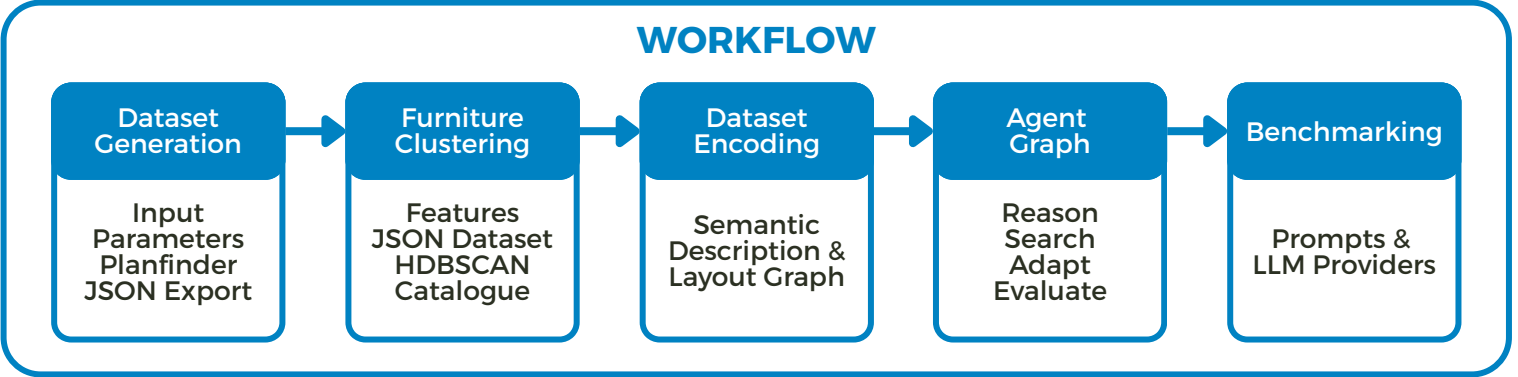
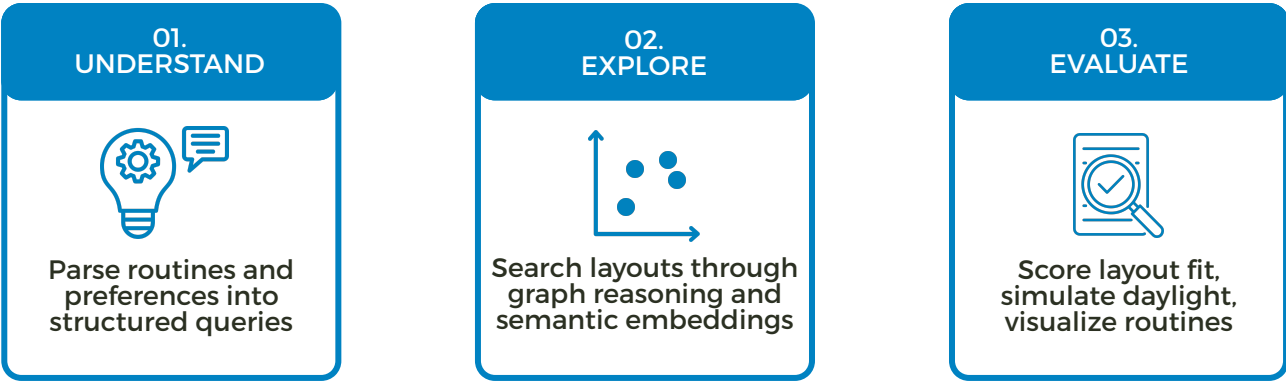
CONTEXT: IAAC Master in Advanced Computational Design (MACAD)

inHabit is an AI spatial intelligence copilot that bridges the gap between architectural layouts and daily routines. Guided by "Describe your day. Find your layout.", it converts natural language text into spatial criteria to aid households, architects, and real estate agents. Powered by LangGraph, Anthropic's API, and a NetworkX spatial graph, it extracts user preferences, runs graph-based semantic searches across 800+ synthetic plans, and overlays sequential daily routines onto layouts with daylight analysis.

inHabit

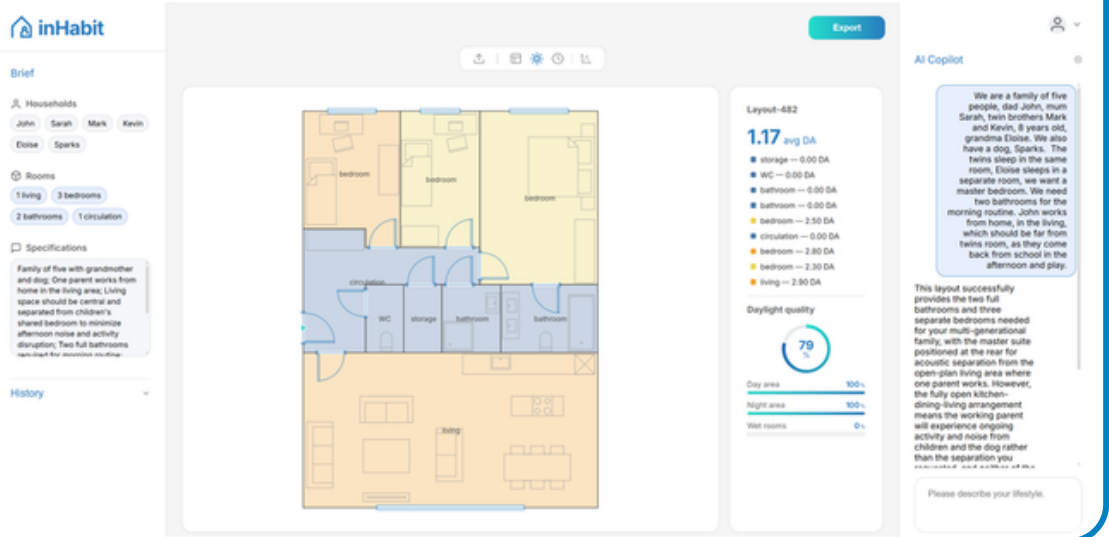
DESCRIBE YOUR DAY. FIND YOUR LAYOUT.

A spatial intelligence copilot that finds the layout that fits the way you live



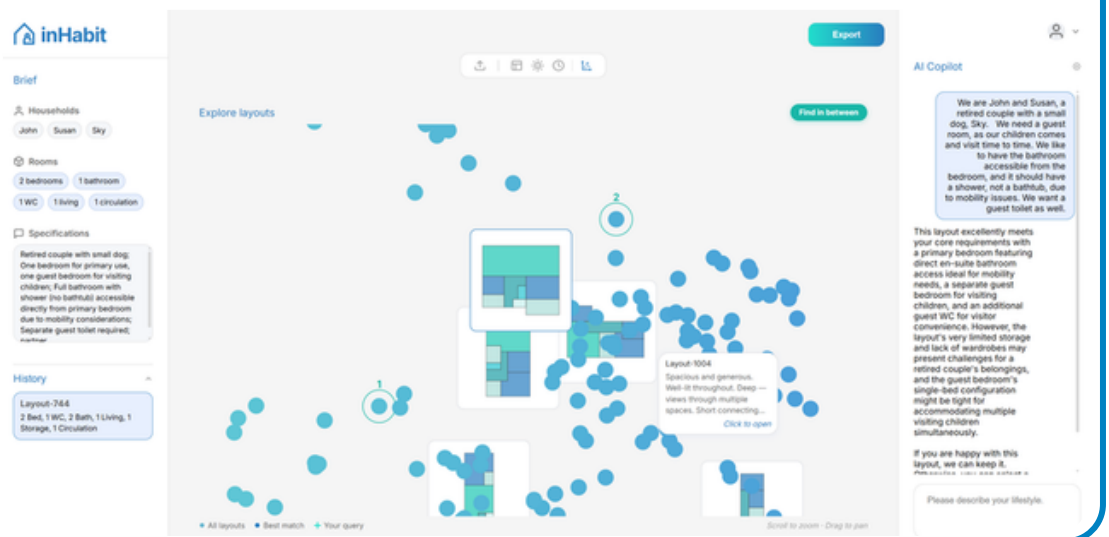
MCP TOOLS

The integration of Python, Grasshopper and Rhino through MCP enables the development of customized tools and analyses within a single pipeline.



GRAPH-BASED DESIGN EXPLORATION

Each layout is encoded as a graph of spatial relationships. The search combines semantic descriptions generated through Anthropic's API with graph-based spatial information.



Brief

- Households
- John
 - Sarah
 - Mark
 - Kevin
 - Eloise
 - Sparks

- Rooms
- 1 living
 - 3 bedrooms
 - 2 bathrooms
 - 1 circulation

Specifications

Family of five with grandmother and dog. One parent works from home in the living area; Living space should be central and separated from children's shared bedroom to minimize afternoon noise and activity disruption. Two full bathrooms required for morning routine.

History



- Layout-482
- 107.10 m²
- storage — 2.4 m²
 - WC — 2.4 m²
 - bathroom — 3.7 m²
 - bathroom — 6.1 m²
 - bedroom — 10.0 m²
 - circulation — 11.0 m²
 - bedroom — 11.1 m²
 - bedroom — 10.7 m²
 - living — 41.8 m²



AI Copilot

We are a family of five people, dad John, mum Sarah, twin brothers Mark and Kevin, 8 years old, grandma Eloise. We also have a dog, Sparks. The twins sleep in the same room, Eloise sleeps in a separate room, we want a master bedroom. We need two bathrooms for the morning routine. John works from home, in the living, which should be far from twins room, as they come back from school in the afternoon and play.

This layout successfully provides the two full bathrooms and three separate bedrooms needed for your multi-generational family, with the master suite positioned at the rear for acoustic separation from the open-plan living area where one parent works. However, the fully open kitchen-dining-living arrangement means the working parent will experience ongoing activity and noise from children and the dog rather than the separation you requested. Good morning! All the best.

Please describe your lifestyle.